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carlin@nyc:~$ ./internship-review
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Beyond the Code

What Building Three AI Products Taught Me

Carlin Hou

Software Engineering Intern

Universal Processing · Summer 2026

\$ whoami



name	Carlin Hou
role	Software Engineering Intern
team	Universal Processing
current	New York City

// background

**My background
combines business and
information systems.**

I look at product decisions from both the user
and engineering sides.

Technical internships taught me to build; startup work taught me to ask who needs it—and why.

My Day Connects New York and Taiwan

// start with context

Before I build, I decide what will move the product forward.

I arrive before 8:30, read Taiwan updates, and set the day's priority.

BEFORE 08:30	sync()	Taiwan updates Email · chat · Jira · priorities
MORNING	build()	Product + engineering Architecture · code · demos · documentation
RECURRING	align()	Cross-team decisions Lewis 1:1 · Taiwan engineers + PM · user-team calls
AFTERNOON	verify()	Test and learn QA · feedback · code review · next steps

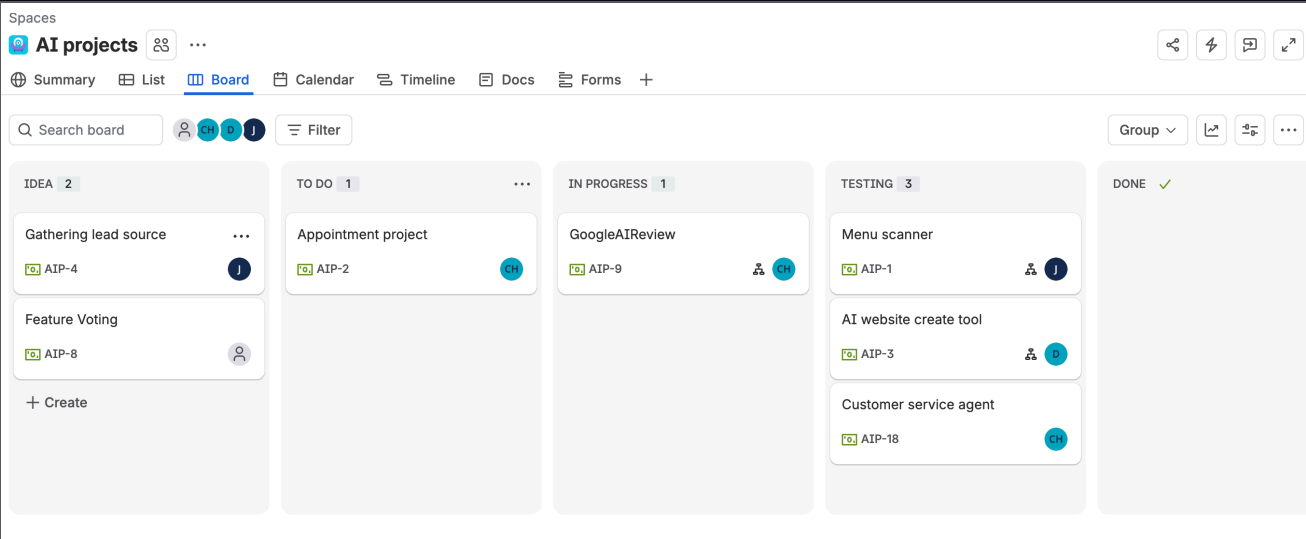
Product decisions → engineering → testing → feedback.

I Turned Unclear Needs Into Buildable Systems

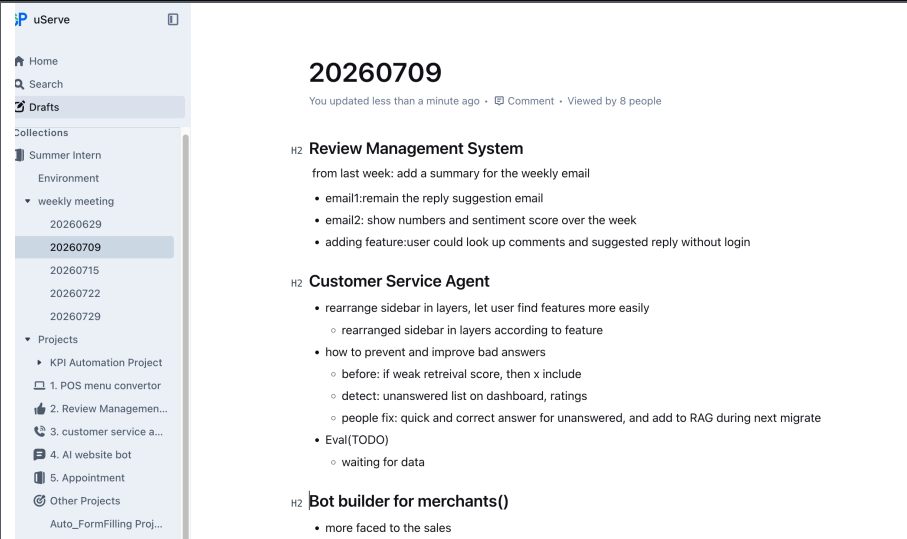
// product loop

Who does the work?	Where does it stall?	What can go wrong?
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workflow |> requirements |> system_design |> tasks |> prototype |> test |> feedback



Jira board · requirements become visible, trackable work



Weekly notes · decisions, feedback, and next steps

— Clear structure keeps engineers—and AI—from building the wrong product faster.

Three Problems. Three Products.

01	<code>respond()</code> Merchant Review Dashboard WORKING · TESTING + REVISION	PROBLEM SOLUTION	Marketing manages Google and Yelp reviews across many merchant accounts. AI drafts in each merchant's voice; sensitive reviews require owner approval.
02	<code>support()</code> 24/7 Support Assistant WORKING · TESTING	PROBLEM SOLUTION	Payment questions arrive after hours; trusted answers are scattered across documents. The assistant searches company knowledge first, then clarifies or hands off.
03	<code>coordinate()</code> Salon Booking Product WORKING END TO END · VALIDATION NEXT	PROBLEM SOLUTION	Booking depends on services, skills, schedules, rooms, and resources—not just an open time. One engine connects natural conversation, availability, and owner controls.

— Automate repetition without removing human judgment.

AI Handles Repetition. People Protect Reputation.

 GBP Reply Manager

Manage reviews

How is any business doing on Google?

Search a business to see its Google rating and a clearly labeled snapshot of recent public reviews. Free, no sign-in.

Manage your reviews

Search

Suggestions update as you type.

Live link: replymanager.userve.io

// problem → solution → value

PROBLEM	Marketing manages Google and Yelp reviews across many accounts, but every merchant has a different voice.
SOLUTION	AI drafts routine replies in that voice; sensitive reviews require owner context and approval.
VALUE	Less account switching, faster routine replies, and safer handling of reputational risk.
collect() → tone_draft() → risk_check() → post_or_approve()	
BUILT	Dashboard · merchant stats · tone-aware drafts · approval flow
TODAY	Working in testing · MKT feedback guides revisions
NEXT	Measure response time · edit rate · approval turnaround

Trusted Answer—or a Clear Human Handoff

// problem → solution → value

PROBLEM Payment problems can happen at any hour, while useful answers are scattered across company documents.

SOLUTION The assistant searches trusted documents first; if unsure, it clarifies and hands off a clearer case.

VALUE Faster first help, less searching, and a better-informed support handoff.

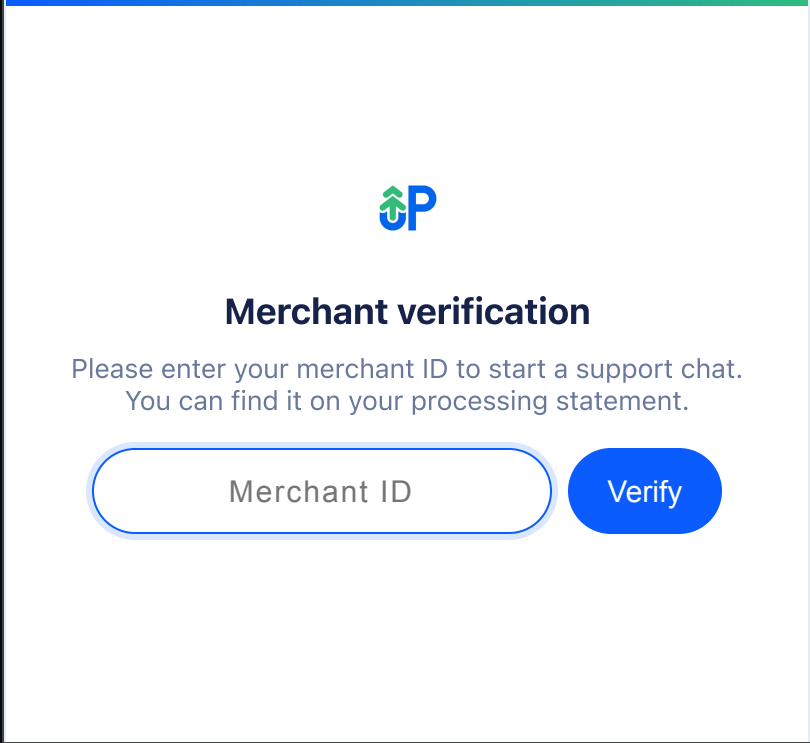
question → trusted_docs → answer_or_clarify → human_handoff

BUILT Structured knowledge · retrieval · clarification · handoff flow

TODAY Working in testing · grounded answers + source checks

NEXT Stronger phone + CRM integration · real support data

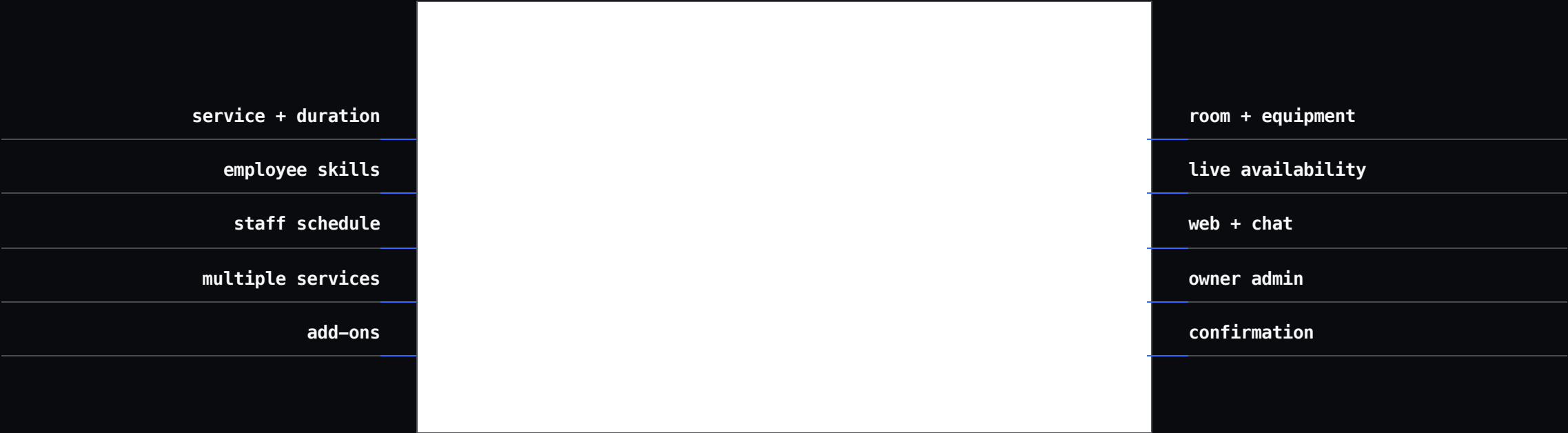
— **What matters is the handoff, not just the chat screen.**



Live link: csagent.userve.io

One Conversation Coordinates the Entire Booking

CUSTOMER	"I need a gel manicure with nail art this Saturday afternoon."	SYSTEM	One engine asks follow-ups, checks real rules, and confirms the booking.
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Working end-to-end product · merchant validation next Use screenshot [Open live app ↗](#)

I owned the rules, matching, availability, admin, and customer experience.

The Bottleneck Moved to Judgment

// the hard questions

01 **Where should automation stop?**

02 **What information should AI trust?**

03 **What happens when AI is uncertain?**

// the boundaries I chose

REVIEWS **People control reputational risk.**

SUPPORT **Trusted documents first; human handoff when uncertain.**

BOOKING **The availability engine controls reservations.**

BEFORE **"Can we write this?"**

→ NOW

"Do we understand it well enough?"

I connect users, business workflows, and technical systems—and own the result.

I began with a narrow idea of engineering.
Now I see the full product loop.

**I want to turn messy
problems into products
people can trust.**

One month left to move these products closer to daily use.

With thanks to Taipei City, Universal Processing, the engineers, coworkers, and interns who shaped this summer.

Thank you.

